BRIDGING 2D EFFICIENCY AND 3D CONTEXT

A MEMORY-GUIDED FRAMEWORK FOR KNEE MRI MULTI-LABEL CLASSIFICATION

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Code on GitHub

Scan to access the website

Main result

ACL AUC 0.951

Meniscus AUC 0.838 on official MRNet sagittal validation.

Key idea

2D speed + 3D context

Depth attention gives slice-level context without a heavy 3D model.

Clinical hook

Memory-guided evidence

Task-specific prototypes and Grad-CAM make predictions discussion-ready.

MOTIVATION

Radiologists inspect knee MRI as a volume, but many automated systems either process slices independently or pay the cost of full 3D models.

Why it is hard

- ACL and meniscus tears can be subtle and sparse.
- Soft tissues often have similar MRI signal intensity.
- Only a few slices may contain decisive evidence.

CONTRIBUTIONS

- Multi-scale center cropping captures both global anatomy and fine lesion texture.
- Depth-aware attention learns which slices should dominate the volume representation.
- Task-aware memory stores diagnostic prototypes separately for ACL and meniscus tears.
- A frozen medical-image ResNet keeps training compact: 650,753 trainable parameters.

DATASETS

Dataset	Train	Val	Role
MRNet	1,130	120	Main benchmark
KneeMRI	641	276	External test

The poster highlights the official MRNet validation results and uses KneeMRI as supporting evidence for external generalization.

IMPLEMENTATION

- Backbone: ResNet-18 initialized with RadImageNet medical pretraining.
- Input: sagittal MRI slices resized to 224 x 224.
- Training: 250 epochs, AdamW, OneCycleLR, mixed precision.
- Frozen backbone keeps optimization focused on memory, projection heads, and classifier.

Augmentation

- Random rotation, horizontal flipping, and affine translation reduce overfitting.
- Frozen backbone keeps the model stable on limited medical labels.

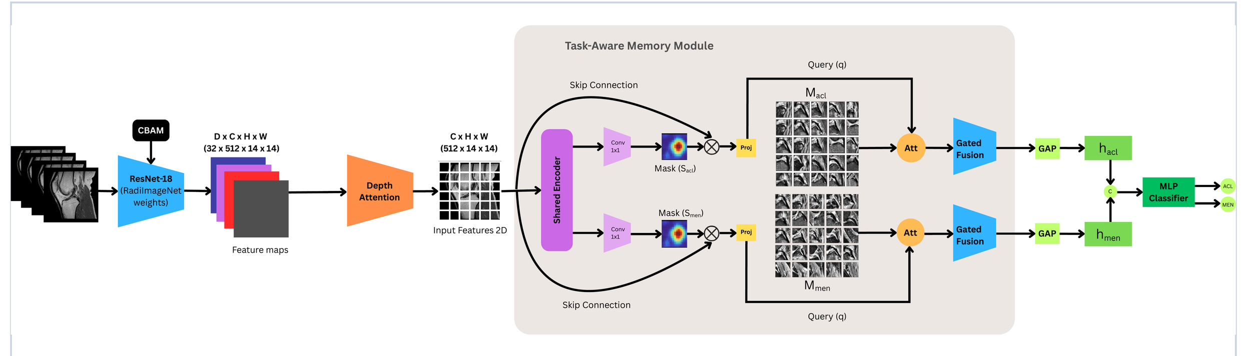
POSTER PITCH

In one sentence

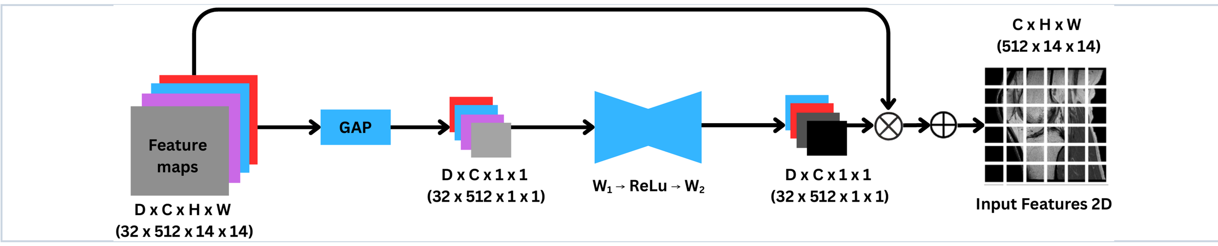
- The model reads MRI slices like a fast 2D network, but uses attention and memory to recover the diagnostic context clinicians need.
- The strongest discussion hook is the link between high ACL AUC and visible lesion localization.

METHOD

SingleViewMemoryNet pipeline

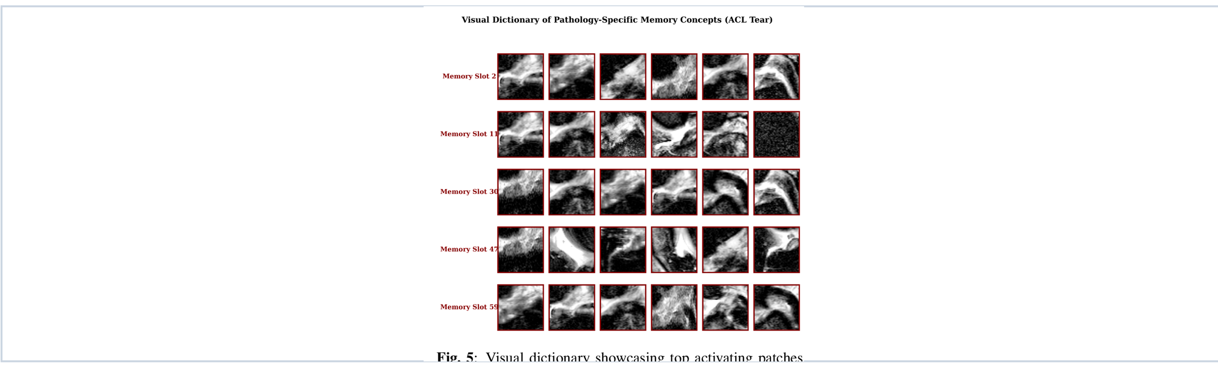


Depth Attention pipeline



Feature maps D x C x H x W → **GAP** slice descriptors → **MLP weights** W1 -> ReLU -> W2 → **Reweight** slice importance → **2D features** C x H x W

TASK-AWARE MEMORY



Memory slots behave like learned visual prototypes. Querying only the most relevant slots helps separate co-occurring diseases before final classification.

EFFICIENCY

Model	Infer.	Total params	Trainable
ELNet	5.98 ms	0.13M	0.13M
Young Seok Jeon	28.25 ms	0.13M	0.13M
Inception V4	17.38 ms	41.08M	41.08M
MRNet	9.41 ms	2.47M	2.47M
Ali Can Kara	13.12 ms	24.70M	1.11M
David Azcona	60.57 ms	11.18M	11.18M
MVGNN	69.07 s	12.49M	0.79M
Ours	**32.24 ms**	**11.83M**	**0.65M**

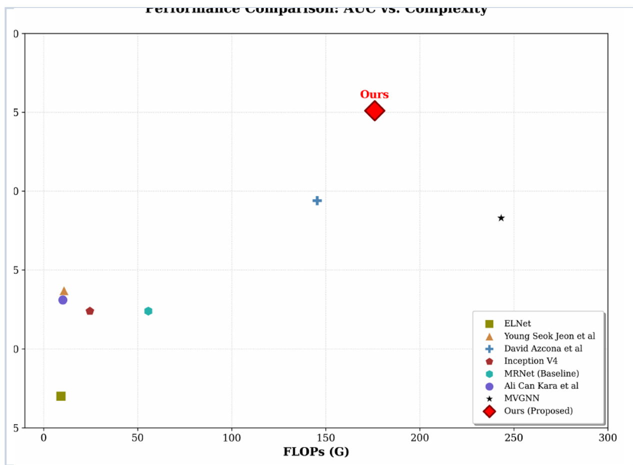
Efficiency takeaway

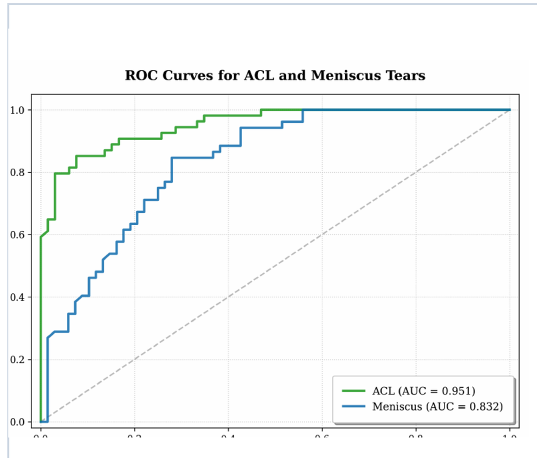
- Ours runs in 32.24 ms per scan, far below MVGNN's 69.07 s.
- It keeps the strongest ACL and meniscus AUC among compared methods.
- Only 0.65M parameters are trained because the medical ResNet is frozen.

RESULTS

Task	AUC	ACC	SE	SP
ACL	0.951	0.902	0.852	0.944
Meniscus	0.838	0.805	0.846	0.765

Performance figures from the paper





COMPARISON

Method	ACL AUC	MEN AUC	Input
MRNet	0.824	0.745	multi-view baseline
ELNet	0.770	0.771	lightweight
MVGNN	0.883	0.802	multi-view
Ali Can Kara	0.895	0.798	prior ACL best
Ours	**0.951**	**0.838**	**single-view**

ABLATION

Configuration	Avg AUC
Baseline ResNet-18	0.850
+ Multi-scale	0.865
+ CBAM	0.872
+ Depth attention	0.888
+ Spatial mask	0.895
Ours: full memory model	**0.925**

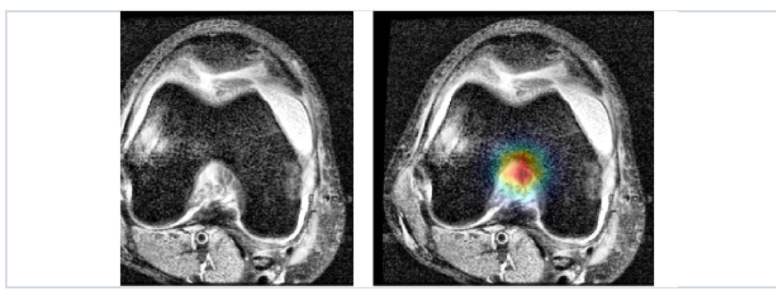
PRETRAINING

Weights	ACL AUC	MEN AUC	Avg
ImageNet	0.901	0.809	0.855
RadImageNet	0.951	0.838	0.895

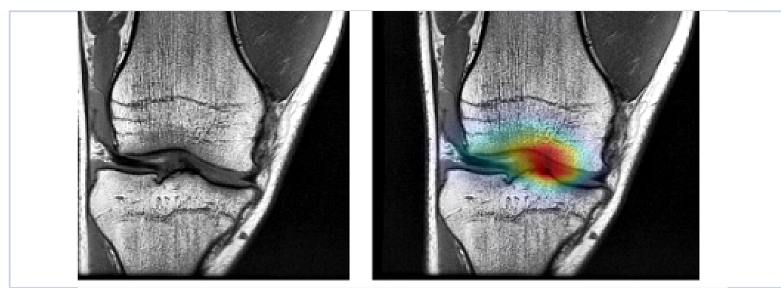
MULTI-PLANE VISUAL EVIDENCE

Sagittal

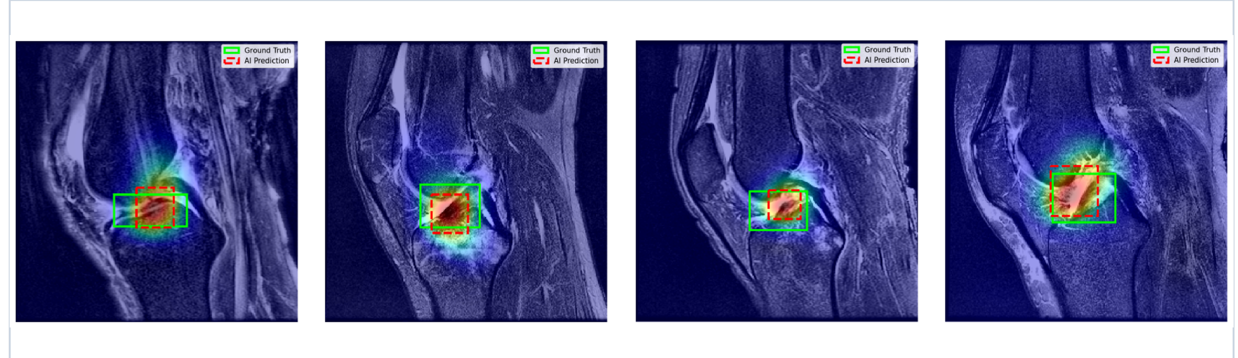
Axial



Coronal



INTERPRETABILITY



The model's attention focuses on clinically meaningful ACL lesion regions in representative validation cases. This turns the output from a black-box score into a discussion-ready diagnostic cue.

GENERALIZATION

Setting	AUC	Sens.	Spec.
Axial ACL	0.9604	0.9259	0.8939
Coronal ACL	0.9537	0.9159	0.9537
KneeMRI ACL	0.9226	0.8326	0.9219
Axial Meniscus	0.8425	0.8077	0.7647
Coronal Meniscus	0.8618	0.9038	0.9091

CONCLUSION

- A single-view 2D backbone can recover useful volumetric context through depth attention.
- Task-specific memory improves separation of ACL and meniscus evidence.
- The proposed model balances accuracy, speed, and interpretability for CAD deployment.

Clinical takeaway

- 32.24 ms per scan keeps the model practical for real-time review.
- Memory and Grad-CAM evidence make predictions easier to question, explain, and discuss.

Next steps

- Extend the module into a multi-view fusion system.
- Validate visual explanations with more clinician annotations.
- Test deployment behavior on prospective clinical data.

Thank you for your attention. Please feel free to ask any questions.